8D report

Complete all fields. Use n/a if not applicable.
Add lines, information and fields as necessary to completely present your problem solving activities.

Version: 2
Issued by: SG / PQA
Date: 30.03.2017

Complaint No.(Q2): 1
Concession No.:

8D started on: 26-12-2024
Intermediate report: 07-01-2025
Final report:

Machine1

Header data

8D-Title:	Motor low performance	SG Material No.:	2004.320.264
Complaint type/mode: 01, 02 and 03-categorie	Performance	Part name:	Carbon Brush
Supplier No.:		SG Production plant:	SEG - Nhp
Supplier name:	AVO Carbon India pvt ltd	Supplier delivery No. / Shipment date :	42424623 28.11.24

Contact Person at SG:	Mr.Raghavendra Ithal	Telephone:	+9180 4103 5264
Business Address:	Nagarathapura ,Bangalore	Email:	Ithal.Raghavendra@seg-
Contact Person at Supplier:	Mrs.Gavithri.N	Telephone:	+919442926570
Business Address:	AVO Carbon India (P) Ltd, Chennai-98.	Email:	gayathri.n@avocarbon.com

D1 Problem Solving Team

Sponsor	Mr.Eipe Thomas	Email	eipe.thomas@avocarbon.com	Phone	+919442926501	Sales and Marketing Director	
Team Leader	Mrs.Gayathri.N	Email	gayathri.n@avocarbon.com.	Phone	+919442926570	Head - Quality	QAD
Team Member	Mr.Sundaresan.R	Email	sundaresan.r@avocarbon.com	Phone	+919442926524	Head - Engineering	ENGG
Team Member	Mr.Suresh.P	Email	suresh.p@avocarbon.com	Phone	+919442926540	Head - Production	Prod
Team Member	Mr.Lakshmanasamv.S	Email	lakshmanasamv@avocarbon.com	Phone	+919442926524	Engineering	ENGG
Team Member	Mr.Saravanan.B	Email	saravanan.b@avocarbon.com	Phone	+919442926573	Quality	QAD
Team Member	Mr.Suresh.A	Email	suresh.a@avocarbon.com	Phone	+919442926573	Quality	QAD

D2 Problem Description

SG description:
Problem description:
Load-1, Load-2 Failure

Date of Detection	26-12-2024
Location of Detection	SEG - Testing
Quantity Complained	56
Quantity Delivered	4200
Quantity Returned	4800
Date of Part Return (leaving SG)	30/12/2024 & 31/12/2024

Failure effect (SG , OEM, FIELD) / :
Motor performance low due to low RPM @ SEG

Supplier description:
Problem description / what, where, when, how much, how often, who, ... etc. /
Load-1, Load-2 Failure

Other SG Parts potentially affected? No *Please specify.*
Other SG Plants potentially affected? No *Please specify.*

Preliminary Risk Assessment Low risk

D3 Containment actions

No.	Part number	Where	Action	Responsible	Introduced on	Control Method	Identification of Controlled	Effective-ness [%]	Quantity blocked TOTAL	Quantity checked TOTAL	Quantity NOK parts
1.	2004.320.264	Avo Stock	100 % Sorting	Avo	29-12-2024	Milli volt drop test	Containment sticker	100%	500	500	Nil
2.	2004.320.264	Rajamane Return	100 % Sorting	Avo	30-12-2024	Milli volt drop test	Containment sticker	100%	1200	1200	Nil
3.	2004.320.264	Rajamane Return	100 % Sorting	Avo	02-01-2025 – 11-01-	Milli volt drop test	Containment sticker	100%	3600	3600	152 - Damage
4.											
5.											
6.											
7.											
8.											
9.											
10.											

Description of Quality Alert and Firewall (until verification of actions in D6).
Parts are undergo 100% containment with milli volt drop testing before dispatch.

D4 Root Cause Analysis

	Technical root cause	Managerial root cause	Date of verification
OCCURENCE	TRC1.: (2.6) Absence of powder in the hopper & hose	MRC1.:No mechanism to indicate the absence of powder	08-01-2025
	TRC2:	...	
NON-DETECTION	TRC1.: (3.2)Layer Missing checks not included in the CP	MRC1.: Layer missing phenomenon not addressed in the PFMEA	07-01-2025

Categorization of failure:
3. Instructions

Reproduction of the defect possible?
The simulation was carried out with the absence of copper powder in the hopper as well as the hose, the machine stopped, and saying thickness was too low and fail in to rejection trap door .When the operator resets the machine again the machine indicates thickness too low and stops. When we adjust the process parameter the machine continue to produced with 10 cycles with the mono layer.This was reproduce the defect as per customer failure part

Expected effects/Risk assessment (Probability):
Risk analysis: */A first risk estimation regarding further affected parts has to be performed, based on the available data: probability of occurrence, probability of detection /*
Based on the data,next batch of suspected lot was found ok with customer testing and brushes are 100% tested with milli volt drop, tested around 5.3k parts passed the milli volt test, so no risk of further production lot... suspected batch Qty - 4200 numbers

D5 Corrective actions and proof of effectiveness:

	Technical root cause	Managerial root cause	Estimated	Effective
OCCURENCE	TRC1.: 1. To place a low-level sensor in the powder hose end that triggers the absence of powder in the hopper and the hose that prevents the layer missing in the product with the indication in the HMI. And the machine will not allow the operator to run the production without pouring the powder in the hopper. 2. After set-up approval Process parameters will be locked and no access given to the operator.	MRC1.: 1. To place a low-level sensor in the powder hose end that triggers the absence of powder in the hopper and the hose that prevents the layer missing in the product with the indication in the HMI. And the machine will not allow the operator to run the production without pouring the powder in the hopper. 2. After set-up approval Process parameters will be locked and no access given to the operator.	100%	27-01-2025
NON-DETECTION	TRC1.: Density check frequency reduced from 2 hrs to 1 hr, in addition to that Brush weight check & Dual Layer confirmation will be added to the in-process check sheet	MRC1.:1. The density check frequency reduced from 2 hrs to 1 hr, in addition to that Brush weight check & Dual Layer confirmation will be added to the in-process check sheet.. 2. Visual standard updated with a dual-layer confirmation 3. Mv drop sample frequency increased from 1no/cavity /basket to 2 nos/cavity/ Basket 4. Brush Weight check added to the Physical property check sheet 5. SOP, PFMEA, CP	100%	07-01-2025

D6 Introduction of corrective actions and tracking of effectiveness

	Technical root cause	Managerial root cause	Effectivene	Responsible	Introduction
OCCURENCE	TRC1.: To place a low-level sensor in the powder hose end that triggers the absence of powder in the hopper and the hose that prevents the layer missing in the product with the indication in the HMI. And the machine will not allow the operator to run the production without pouring the powder in the hopper. 2. After set-up approval Process parameters will be locked and no access given to the operator.	MRC1.: 1. To place a low-level sensor in the powder hose end that triggers the absence of powder in the hopper and the hose that prevents the layer missing in the product with the indication in the HMI. And the machine will not allow the operator to run the production without pouring the powder in the hopper. 2. After set-up approval Process parameters will be locked and no access given to the operator.		Avo	27-01-2025
	TRC 2:			Avo	07-01-2025
NON-DETECTION	TRC1. Density check frequency reduced from 2 hrs to 1 hr, in addition to that Brush weight check & Dual Layer confirmation will be added to the in-process check sheet	MRC1.: 1. The density check frequency reduced from 2 hrs to 1 hr, in addition to that Brush weight check & Dual Layer confirmation will be added to the in-process check sheet.. 2. Visual standard updated with a dual-layer confirmation 3. Mv drop sample frequency increased from 1no/cavity /basket to 2 nos/cavity/ Basket 4. Brush Weight check added to the Physical property check sheet 5. SOP, PFMEA, CP		Avo	07-01-2025

Removal of containment actions:
Based on effectivenss of corrective action we will plan to stop the containment action, till we continue the effective containment.

removed on: Feb/2024

D7 Prevention of recurrence of the non-conformity
(FMEA, directives, PQP, drawing, ... etc.):
- To place a low-level sensor in the powder hose end that triggers the absence of powder in the hopper and the hose that prevents the layer missing in the product with the indication in the HMI
- Powder mix bags are to be introduced for one mix to prevent pouring same mix powder in the both hopper.

Responsible: due date: completed on: 27-01-2025

Lessons Learned transfer to other Processes, Products, Plants
Could the root cause affect other processes, products or sites? ..Which?
All actions are to be transfer to other applicable similar machines.

D8 Final Meeting

Participants: Avo Team, Rajamane and SEG Team
Accomplished on: 27-01-2025
Comments:

Signatures

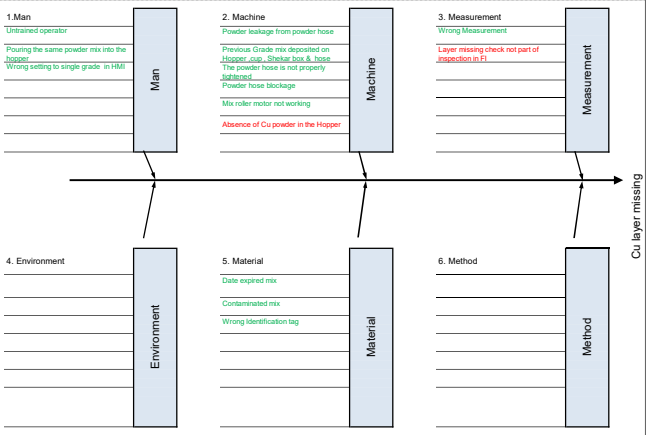
Plant manager:	Name: Mr.Sridhar	Date: 27-01-2025	Signature:
Sponsor:	Name: Mr.Eipe Thomas	Date: 27-01-2025	Signature:
Team Leader:	Name: Mrs.Gayathri.N	Date: 27-01-2025	Signature:

D2 - Failure Visualization

Specific requirement:
Copper rich Grade (OK)

Actual value (nonconformity):
Graphite rich Grade (NOK)

D4 - Ishikawa



Use following colors to categories the root causes in the Ishikawa:

Excluded cause | Possible cause | Most probable cause

Reason of exclusion:
1.1- Verified with in process records found machine operated by experienced regular operator run the machine (Emp 5066,5047) they are well trained and experienced operators.
1.2 The simulation was carried out with the same powder in the two hoppers and the machine stopped because the thickness measured was too low in the HMI. There is a control in the Hopper colour, Powder storage colour, and grade sticker on the hopper and every two hours there is a density check in place hence so the possibility of occurrence will be less.

D4 - 5 Why?

The problem with which the 5Why question chain is started:
Absence of Cu powder in the Hopper &hose

The problem with which the 5Why question chain is started:
Layer missing check not part of inspection in FI

	Occurrence		Non-Detection
1st Why?	Q: Why brush getting crack A: Copper Rich mix not filled	1st Why?	Q: Cu layer missing A:Layer confirmation check not included in the inprocess check sheet
2nd Why?	Q: Whv Copper Rich mix not filled A: Copper Powder missing in the hopper and the hose	2nd Why?	Q: Whv Layer confirmation check not included in the inprocess check sheet A: Not included in the CP
3rd Why?	Q: Why Copper Powder missing in the hopper and the hose A: Operator missed to pour the copper powder in the hopper, and the machine indicates the thickness too low and stops . When the operator modify the process parameter the machine continued produce with mono layer	3rd Why?	Q: Why Not included in the CP A:This phenomena not covered in the PFMEA
4th Why?	Q: Whv Operator missed to pour the copper powder in the hopper, and the A: There is no trigeering system genertaed to pour the powder in to the hopper and also there is no locking system provided for the process parameter	4th Why?	Q: This phenomena not covered in the PFMEA A:we check the thickness of brush while adjusting the process parameter not the layer confirmation check.
5th Why?	Q: There is no trigeering system genertaed to pour the powder in to the hopper and also there is no locking system provided for the process parameter A:	5th Why?	Q: A:

The problem with which the 5Why question chain is started:

The problem with which the 5Why question chain is started:

	Occurrence		Non-Detection
1st Why?		1st Why?	
2nd Why?		2nd Why?	
3rd Why?		3rd Why?	
4th Why?		4th Why?	
5th Why?		5th Why?	

The problem with which the 5Why question chain is started:

The problem with which the 5Why question chain is started:

	Occurrence		Non-Detection
1st Why?		1st Why?	Q: A:
2nd Why?		2nd Why?	Q: A:
3rd Why?		3rd Why?	Q: A:
4th Why?		4th Why?	Q: A:
5th Why?		5th Why?	Q: A:

The problem with which the 5Why question chain is started:

The problem with which the 5Why question chain is started:

	Occurrence		Non-Detection
1st Why?	Q: A:	1st Why?	Q: A:
2nd Why?	Q: A:	2nd Why?	Q: A:
3rd Why?	Q: A:	3rd Why?	Q: A:
4th Why?	Q: A:	4th Why?	Q: A:
5th Why?	Q: A:	5th Why?	Q: A: